Module 05 — File Handling and Logging

Phase: Advanced Python
Previous module: Exception Handling
Next module: Git and GitHub

Module Outcome:

- The problem — RAM forgets, files remember
- Types of files — text vs binary
- Opening a file — `open()` and `with open()`
- File modes — read, replace, append (`r` / `w` / `a`)
- Writing and reading text
- Handling missing files — `FileNotFoundError`
- CSV for tables, JSON for config
- Logging — levels and file handlers
- Advanced awareness — cursor, binary, pickle, directories

Your Python program forgets everything when it stops. This module is how you make it remember.

Last module, a missing file raised `FileNotFoundError`. You could catch it. You could not yet **create** the file, **append** a row, or **read** last month's expenses back. That is the job now.

The problem: where does data go?

Definition: A **file** is data stored on disk. It is still there after the program closes.
Purpose: Variables live in RAM. When the program ends, that RAM is released. Student records, expense rows, and settings cannot live only in a list.

Before this module

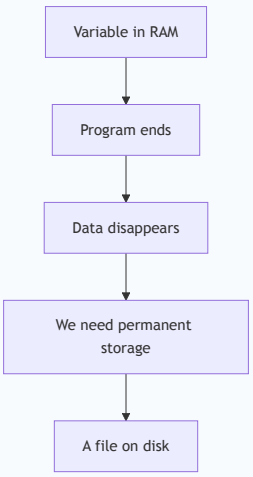
```
expenses = []
expenses.append("Tea ₹120")
# Program stops → that list is gone.
```

```
Python running  →  expenses = ["Tea ₹120"]   (RAM)
Python stops    →  💥 data gone
```

After this module

```
Python application
       │
       ├── data/expenses.csv     (the rows)
       ├── data/config.json      (categories, daily limit)
       └── logs/app.log          (what happened)
```

```
Python stops    →  data remains ✅
```



Key takeaways:

- RAM dies with the process; a **file** survives
- This module is one story: **make Python remember**
- We will build toward one app: an **AcmeHR expense tracker**

Where used?

- **AcmeHR** / payroll — append expense rows, export a CSV, keep a daily limit in config
- **TechMart** — import a product table, load `config.json` at startup
- Attendance and audit — append-only text that survives a restart
- Servers — `logging` to `app.log` when nobody is watching the terminal
- **MedCare** / **SecureBank** — never log passwords, OTPs, or card numbers

We will use the **same** three paths in almost every demo:

```
expense_tracker/
│
├── data/
│   ├── expenses.txt    ← first: plain text
│   ├── expenses.csv    ← then: a table
│   └── config.json     ← settings
│
└── logs/
    └── app.log         ← what the app did
```

You are not collecting random filenames. You are growing **one** application.

What you must remember vs what you only need to recognise

Not every API in this file is equal. Use this map while you read.

Must know (tonight's tracker): `open()`, `with open()`, modes `r` / `w` / `a`, `write()`, `read()`, `readline()`, `for line in f`, CSV, JSON, `logging`, `FileNotFoundError`.

Should know (you will see these): `readlines()`, `read(n)`, `tell()` / `seek()`, `Path.exists()`, binary `rb` / `wb`, `os.makedirs()`.

Awareness (interview / later): `r+` `w+` `a+` `x`, pickle, `os.walk()`, `RotatingFileHandler`.

Types of files

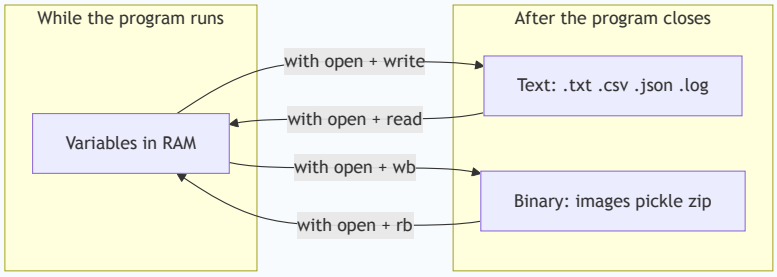
There are **two** kinds. They are different worlds.

```
TEXT                            BINARY
----------------                ----------------
Characters                      Bytes
"Hello"                         b"\x89PNG..."
```

```
encoding="utf-8"                no encoding
modes: r / w / a                modes: rb / wb / ab
```

1. Text files — character data you can open in Notepad: `.txt`, `.csv`, `.json`, `.log`.
2. Binary files — bytes: images, zip, pickle. Do **not** open a `.jpg` in text mode.

Tonight's tracker is **text** (CSV, JSON, logs). Binary is in the Advanced section — know that it exists; do not mix the two.



Key takeaways:

- **Text** = characters (always set `encoding="utf-8"`)
- **Binary** = bytes (`rb` / `wb`) — no encoding
- Do not use text mode for a `.jpg` or other arbitrary bytes

Important Concepts

1. Opening a file — `open()` and `with open()`

Definition: Before any read or write, the file must be **opened**. `open()` returns a file object.
Purpose: Opening a file borrows a **system resource** (an OS handle). When you are finished, that resource should be released.

```
f = open("data/expenses.txt", "w", encoding="utf-8")
f.write("2026-09-04 | food | 120 | Tea\n")
f.close()
```

`f.close()` flushes leftover data and releases the handle.

But remembering `close()` is easy to forget — especially when an exception fires in the middle. That is why Python gives us `with`. The file closes when the block ends, even if the last module's `ValueError` shows up.

From today, this is the habit:

```
with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | food | 120 | Tea\n")
# file is already closed here
```

Note:
- Manual `f = open(...)` / `f.close()` is the *reason* `with` exists — it is not the style you practise.
- Always pass `encoding="utf-8"` for text. Skipping it → `UnicodeDecodeError` on some Windows machines.

Remember: Prefer `with open(...) as f:` — the file closes even if an exception fires inside the block.

Create the `data/` folder, write one AcmeHR expense line with `with open`, then read it back and print it.

```
import os

os.makedirs("data", exist_ok=True)
```

```
with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | food | 120 | Tea\n")

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

Expected Output:

```
2026-09-04 | food | 120 | Tea
```

Inside `with open`, write a line then raise `ValueError`. After catching it, reopen the file — it closed cleanly, so a new `with` works.

```
import os

os.makedirs("data", exist_ok=True)

try:
    with open("data/expenses.txt", "w", encoding="utf-8") as f:
        f.write("partial row\n")
        raise ValueError("deliberate crash after write")
except ValueError as e:
    print("Caught:", e)

with open("data/expenses.txt", "a", encoding="utf-8") as f:
    f.write("appended after crash\n")

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

Expected Output:

```
Caught: deliberate crash after write
partial row
appended after crash
```

That is the bridge from Exception Handling: `with` is the `finally` for files, without writing `finally` yourself.

2. File modes — read, replace, or append

Definition: The **mode** is *why* you opened the file.
Purpose: The wrong mode silently wipes a month of expenses (`w`) or refuses to open (`r` on a missing file).

For this course you need **three** modes first:

```
What do I want?

                    File
                     |
          ┌──────────┼──────────┐
          ↓          ↓          ↓
        Read      Replace      Add
          ↓          ↓          ↓
          r          w          a
```

Mode	Plain English	Missing file?	Existing content?
`r`	I want to **read** what is already there	`FileNotFoundError`	Left as-is
`w`	I want to **create / replace** the file	Creates it	**Cleared** (truncated)
`a`	I want to **add** without deleting the old lines	Creates it	Kept; new writes go at the **end**

`r` is the **default** if you omit the mode.

Python has other modes (`r+`, `w+`, `a+`, `x`). We will name them in Advanced. You do not need them to build the tracker.

Common Error: Opening with `"w"` when you meant `"a"` — `w` empties the file before you write.

Pause. Do not run yet. What will `data/expenses.txt` contain after this program?

```
import os

os.makedirs("data", exist_ok=True)

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | food | 120 | Tea\n")

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | travel | 50 | Metro\n")
```

A. Both lines (Tea and Metro)
B. Only Metro
C. Only Tea
D. Error

Now run it, then read the file.

```
import os

os.makedirs("data", exist_ok=True)

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | food | 120 | Tea\n")

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | travel | 50 | Metro\n")

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    print(repr(f.read()))
```

Expected Output:

```
'2026-09-04 | travel | 50 | Metro\n'
```

The answer is **B**. The second `"w"` **replaced** the file. Tea is gone.

Remember: `w` means replace. For a running log or a growing expense list, you almost always want `a`.

Write the first expense with `w` (create the file), append two more with `a`, then read so all three lines remain.

```
import os

os.makedirs("data", exist_ok=True)

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | food | 120 | Tea\n")

with open("data/expenses.txt", "a", encoding="utf-8") as f:
    f.write("2026-09-04 | travel | 50 | Metro\n")
    f.write("2026-09-04 | food | 40 | Samosa\n")

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

Expected Output:

```
2026-09-04 | food | 120 | Tea
2026-09-04 | travel | 50 | Metro
2026-09-04 | food | 40 | Samosa
```

Note:

- `w` is the **mode**. `write()` is the **method**. Opening with `w` does not write anything by itself.
- `r` is the mode. `read()` is the method.
- `write()` does **not** add `\n` for you. Omit it → the next write sticks to the same line.
- `writelines(list)` writes a list of strings. It also does **not** add `\n`. Prefer a loop with `write(line + "\n")`.

3. Reading data from text files

Definition: After you open for read, you pull characters (or lines) from the file.

Purpose: The tracker has to show last week's expenses, not only the ones still in RAM.

Method	Returns	When to use
`read()`	Entire file as **one string**	Small files
`read(n)`	Next `n` characters	Peek / should-know
`readline()`	**One** line	Line-by-line protocols
`readlines()`	**List** of all lines	Medium files — whole list in RAM
`for line in f`	One line at a time	**Large files** — this is the habit

What if the file is huge? A 2 GB server log must not be loaded with `read()`. Iterate:

```
with open("logs/app.log", "r", encoding="utf-8") as f:
    for line in f:
        ...
```

The file pointer moves as you read. A second `read()` without reopening (or `seek` — Advanced) often returns empty.

Note:
- `readlines()` vs `for line in f`: the first builds a **list in memory**; the second yields **one line at a time**.
- Reading a huge file with `read()` → `MemoryError`. Iterate.

Write three expense lines, then print each line with `for line in f` (strip the newline).

```
import os

os.makedirs("data", exist_ok=True)
lines = [
    "2026-09-04 | food | 120 | Tea",
    "2026-09-04 | travel | 50 | Metro",
    "2026-09-04 | food | 40 | Samosa",
]

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    for line in lines:
        f.write(line + "\n")

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    for line in f:
        print(line.strip())
```

Expected Output:

```
2026-09-04 | food | 120 | Tea
2026-09-04 | travel | 50 | Metro
2026-09-04 | food | 40 | Samosa
```

On the same small file, show `read()` (whole string) next to iterating lines — two ways to read, different memory behaviour.

```
import os

os.makedirs("data", exist_ok=True)

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("Tea\nMetro\nSamosa\n")

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    all_text = f.read()  # OK for small files only
    print("read():", repr(all_text))

with open("data/expenses.txt", "r", encoding="utf-8") as f:
    for line in f:  # prefer this for large logs
        print("line:", line.strip())
```

Expected Output:

```
read(): 'Tea\nMetro\nSamosa\n'
line: Tea
line: Metro
line: Samosa
```

4. Handling missing files

Definition: A path might not exist yet. `open(..., "r")` then raises `FileNotFoundError`.
Purpose: A missing CSV should print a clear message on the clerk's screen, not a traceback.

This is last module's exception, used on purpose.

```
try:
    with open(path, "r", encoding="utf-8") as f:
        ...
except FileNotFoundError:
    print("No expenses yet — starting fresh.")
```

`pathlib.Path.exists()` asks first (Look Before You Leap). After Exceptions week, `try` / `except FileNotFoundError` is the usual Python style (EAFP).

Note:
- Both styles are valid. Do not skip handling.
- `"w"` / `"a"` *create* a missing file — you do not get `FileNotFoundError` in those modes.

Try to read `data/missing.txt` (it is not there), catch `FileNotFoundError`, then read `data/expenses.txt` which you just wrote.

```
import os

os.makedirs("data", exist_ok=True)

with open("data/expenses.txt", "w", encoding="utf-8") as f:
    f.write("2026-09-04 | food | 120 | Tea\n")

def read_text(path):
    try:
        with open(path, "r", encoding="utf-8") as f:
            return f.read()
    except FileNotFoundError:
        print(f"File not found: {path}")
        return None

for fname in ("data/expenses.txt", "data/missing.txt"):
    print("Checking:", fname)
    data = read_text(fname)
    if data is not None:
        print(data.strip())
```

Expected Output:

```
Checking: data/expenses.txt
2026-09-04 | food | 120 | Tea
Checking: data/missing.txt
File not found: data/missing.txt
```

5. CSV — when the data is a table

Definition: CSV means **Comma Separated Values** — a text table Excel and Sheets understand.
Purpose: `Tea | 120` as free text is fine for two lines. A month of expenses needs **columns**: date, category, amount, note.

Python's `csv` module:

- `csv.writer` / `csv.reader` — rows as lists
- `csv.DictWriter` / `csv.DictReader` — rows as dicts keyed by the header (prefer this when columns have names)

Note:
- Open CSV with `newline=""` (Python 3 / Windows extra blank lines otherwise).
- Do not open CSV as `"wb"`.
- Amounts in CSV are **strings** until you convert: `int(row["amount"])`.

Write `data/expenses.csv` with `DictWriter`, then read it back and print the total amount.

```
import csv
import os

os.makedirs("data", exist_ok=True)

rows = [
    {"date": "2026-09-04", "category": "food", "amount": "120", "note": "Tea"},
    {"date": "2026-09-04", "category": "travel", "amount": "50", "note": "Metro"},
    {"date": "2026-09-04", "category": "food", "amount": "40", "note": "Samosa"},
]

with open("data/expenses.csv", "w", newline="", encoding="utf-8") as f:
    writer = csv.DictWriter(f, fieldnames=["date", "category", "amount", "note"])
    writer.writeheader()
    writer.writerows(rows)

total = 0
with open("data/expenses.csv", newline="", encoding="utf-8") as f:
    for row in csv.DictReader(f):
        print(row["note"], "₹" + row["amount"])
        total += int(row["amount"])
print("Total:", total)
```

Expected Output:

```
Tea ₹120
Metro ₹50
Samosa ₹40
Total: 210
```

Interview: `DictReader` uses the header row as keys. `reader` gives lists — you remember column indexes yourself.

Plain `expenses.txt` was the learning wheel. From here, the tracker's data file is **`data/expenses.csv`**.

6. JSON — when the data is config (or nested)

Definition: JSON is nested data as text — the language of APIs and config files.
Purpose: Categories, daily limit, and currency belong in `data/config.json`, not hardcoded in the middle of `main`.

Function	Reads / writes
`json.dump(obj, file)`	Object → **file**
`json.load(file)`	**File** → object
`json.dumps(obj)`	Object → **string** (`s` = string)
`json.loads(s)`	**String** → object

```
Is my data basically a table?
        |
       YES  →  CSV
        |      Excel / report / import / export

Is my data configuration or a nested structure?
        |
       YES  →  JSON
               API / config / preferences
```

Write `data/config.json` for the expense tracker, load it, and print the daily limit.

```
import json
import os

os.makedirs("data", exist_ok=True)

config = {
    "app_name": "ExpenseTracker",
    "currency": "INR",
    "max_daily_limit": 5000,
    "categories": ["food", "travel", "rent"],
}

with open("data/config.json", "w", encoding="utf-8") as f:
    json.dump(config, f, indent=2)

with open("data/config.json", "r", encoding="utf-8") as f:
    loaded = json.load(f)
    print("Daily limit:", loaded["max_daily_limit"])
    print("Categories:", loaded["categories"])
```

Expected Output:

```
Daily limit: 5000
Categories: ['food', 'travel', 'rent']
```

Remember: `json.load` → file. `json.loads` → string. The extra **s** is "string".

7. Logging — when nobody is watching the terminal

Definition: Logging writes messages with a **level**, a timestamp, and a destination (console, file, or both).
Purpose: `print` is for learning. A real application runs on a server. Nobody is staring at that terminal.

A clerk says: **"My expense wasn't saved."** You need answers:

```
When did it happen?
What operation was running?
Was it a warning, or did something fail?
```

```
print()     →  good for class / quick debug
logging     →  good for real applications
```

Remember the meaning, not the numbers.

Level	Think of it as	Expense Tracker example
DEBUG	Developer detail	Reading row 25
INFO	Normal event	Expense added
WARNING	Something unusual	Daily limit almost reached
ERROR	Something failed	CSV could not be read
CRITICAL	Serious failure	Application cannot start

The numbers (you will see them in docs): DEBUG=10, INFO=20, WARNING=30, ERROR=40, CRITICAL=50. A logger set to INFO **hides** DEBUG.

```
import logging
logging.basicConfig(level=logging.INFO, format="%(asctime)s %(levelname)s %(message)s")
logging.info("Expense added: ₹120 Tea")
logging.error("Could not open expenses.csv")
```

Note:
- `print` in production → no levels, no timestamps, gone when the console is gone. Use `logging`.
- Configure logging in `main`, not at import time inside a library.
- `basicConfig` runs **once** per process. Restart the interpreter if a second call seems to do nothing.

Interview: `print` is for quick demos; `logging` gives levels, timestamps, and file/console handlers you can filter.

Configure logging at INFO. Log DEBUG, INFO, and ERROR so DEBUG is filtered out.

```
import logging

logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")

logging.debug("Reading row 25 — this will NOT appear at INFO level")
logging.info("Expense added: ₹120 Tea")
logging.warning("Daily limit almost reached")
logging.error("Failed to parse amount: 'abc'")
```

Expected Output:

```
INFO Expense added: ₹120 Tea
WARNING Daily limit almost reached
ERROR Failed to parse amount: 'abc'
```

Attach a file handler (`logs/app.log`) and a console handler on logger `expense` , then log INFO and ERROR to both.

```
import logging
import os

os.makedirs("logs", exist_ok=True)

logger = logging.getLogger("expense")
logger.setLevel(logging.DEBUG)
logger.handlers.clear()

fmt = logging.Formatter("%(asctime)s %(levelname)s %(message)s")

file_handler = logging.FileHandler("logs/app.log", encoding="utf-8")
file_handler.setFormatter(fmt)
stream_handler = logging.StreamHandler()
stream_handler.setFormatter(fmt)

logger.addHandler(file_handler)
logger.addHandler(stream_handler)

logger.info("Expense tracker started")
logger.error("Could not open data/expenses.csv")
print("Also check logs/app.log on disk")
```

You should see the same INFO and ERROR lines on the console **and** inside `logs/app.log` .

Awareness: When a log file grows forever, production uses `RotatingFileHandler` (size) or `TimedRotatingFileHandler` (time). You do not need that for tonight's tracker.

Note: Never log passwords, OTPs, full card numbers, Aadhaar, or session tokens (GDPR / India DPDP). Log that an auth attempt failed, not the secret.

---

8. Advanced file handling (awareness)

You do **not** need this section to finish tonight's tracker. Skim once. Come back when a job (or an interview) names the tool.

File cursor — `tell()` and `seek()`

Think of the file as a line of characters and a **cursor**:

```
FILE   2 0 2 6 - 0 9 - 0 4 ...
       ↑
     cursor
```

- `tell()` — "Where am I?" (index from the start; first character is **0**)
- `seek(offset)` — "Move me there."

For **text** files in Python 3, `seek` is used from the **beginning**. Passing a random non-zero offset that did not come from `tell()` can error. Binary files can also seek from the current position or the end.

In-place overwrite needs a read-and-write mode (`r+`) and replacement text that fits the gap.

Write a status line, read it, `seek` to `PENDING`, overwrite it, then read from the start again.

```
data = "AcmeHR status: PENDING"
with open("data/status.txt", "w", encoding="utf-8") as f:
    f.write(data)

with open("data/status.txt", "r+", encoding="utf-8") as f:
    print(f.read())
    print("Cursor after read:", f.tell())
    f.seek(15)  # start of PENDING
    print("Cursor after seek:", f.tell())
    f.write("CLEARED")
    f.seek(0)
    print(f.read())
```

Expected Output:

```
AcmeHR status: PENDING
Cursor after read: 22
Cursor after seek: 15
AcmeHR status: CLEARED
```

Interview: `tell()` reports where you are; `seek()` moves there.

Binary files

Text and binary are different worlds (see the box at the top). Open with `rb` / `wb` / `ab`. Do **not** pass `encoding`.

The lesson to keep: **do not use text mode for arbitrary binary data** (a `.jpg` is not UTF-8 text).

Write bytes to `data/badge.bin`, copy them to `data/badge_copy.bin`, and print whether the copy matches.

```
import os

os.makedirs("data", exist_ok=True)
original = b"PNG-demo-bytes-not-a-real-image"
```

```
with open("data/badge.bin", "wb") as f:
    f.write(original)

with open("data/badge.bin", "rb") as src, open("data/badge_copy.bin", "wb") as dst:
    dst.write(src.read())

with open("data/badge_copy.bin", "rb") as f:
    copied = f.read()

print("Copy matches:", copied == original)
```

Expected Output:

```
Copy matches: True
```

The same pattern copies a real `.jpg`: `open("photo.jpg", "rb")` → `open("newpic.jpg", "wb")`.

Other modes — `r+`, `w+`, `a+`, `x`

Mode	Meaning	Use when
`r+`	Read and write; file must exist; no truncate	In-place edit (`seek` example above)
`w+`	Write and read; **truncates** like `w`	Rare; easy to wipe data
`a+`	Append and read	Read a log you are also appending
`x`	Create exclusive; error if the file exists	"Create only if new"

Suffix `b` for binary: `rb`, `wb`, `ab`, `r+b`, …

Remember: If you are unsure, you still only need `r`, `w`, and `a`.

Pickling and unpickling

Pickling writes a Python object's state as bytes (`pickle.dump`). **Unpickling** reads it back (`pickle.load`). Use **binary** mode (`wb` / `rb`).

Do not unpickle data you do not trust — pickle can run code.

For tables and config you share with humans or other languages, prefer **JSON** or **CSV**.

Pickle an AcmeHR employee dict to `data/emp.dat`, then unpickle and print name and salary.

```
import os
import pickle

os.makedirs("data", exist_ok=True)
employee = {"eno": 101, "ename": "Anita", "esal": 45000, "eaddr": "Hyderabad"}

with open("data/emp.dat", "wb") as f:
    pickle.dump(employee, f)

with open("data/emp.dat", "rb") as f:
```

```
    obj = pickle.load(f)
    print("After unpickling:", obj["ename"], obj["esal"], obj["eaddr"])
```

Expected Output:

```
After unpickling: Anita 45000 Hyderabad
```

To write several objects, `dump` in a loop. To read them back, `load` in a loop until `EOFError`.

Directories

You already used **`os.makedirs(path, exist_ok=True)`** to create `data/` and `logs/`. That is the should-know piece.

Need	Function
Current folder	`os.getcwd()`
Create nested folders	`os.makedirs(path, exist_ok=True)`
List one folder	`os.listdir(path)` — **not** nested files
List folder **and** subfolders	`os.walk(path)` — awareness

Interview: `listdir()` is one directory. `walk()` yields `(dirpath, dirnames, filenames)` for that directory **and** every subdirectory.

Print the current folder, ensure `data/` exists, then `walk` it (names on your machine will match what you have created).

```
import os

print("Current Working Directory:", os.getcwd())
os.makedirs("data", exist_ok=True)

for dirpath, dirnames, filenames in os.walk("data"):
    print("Path:", dirpath)
    print("Dirs:", dirnames)
    print("Files:", filenames)
```

Remember: Don't confuse these — `w` is a **mode**, `write()` is a **method**. `r` is a mode, `read()` is a method. `json.load` reads a **file**; `json.loads` reads a **string**. `readlines()` builds a list in RAM; `for line in f` yields one line at a time.

Step-by-Step: one assemble — AcmeHR expense slice

Type this with the class. It stitches **today's** must-knows on **one** product: folders, `with`, UTF-8, CSV `newline=""`, `DictWriter` / `DictReader`, JSON config, `logging` to `logs/app.log`. The overnight assignment is the **full** tracker kit — same product, more complete ticket.

```
import csv
import json
```

```
import logging
import os

os.makedirs("data", exist_ok=True)
os.makedirs("logs", exist_ok=True)

logging.basicConfig(
    filename="logs/app.log",
    level=logging.INFO,
    format="%(levelname)s %(message)s",
)

config = {
    "app_name": "ExpenseTracker",
    "currency": "INR",
    "max_daily_limit": 5000,
    "categories": ["food", "travel", "rent"],
}
with open("data/config.json", "w", encoding="utf-8") as f:
    json.dump(config, f, indent=2)

rows = [
    {"date": "2026-09-04", "category": "food", "amount": "120", "note": "Tea"},
    {"date": "2026-09-04", "category": "travel", "amount": "50", "note": "Metro"},
]
with open("data/expenses.csv", "w", newline="", encoding="utf-8") as f:
    writer = csv.DictWriter(f, fieldnames=["date", "category", "amount", "note"])
    writer.writeheader()
    writer.writerows(rows)
logging.info("Wrote data/expenses.csv")

total = 0
with open("data/expenses.csv", newline="", encoding="utf-8") as f:
    for row in csv.DictReader(f):
        print(f"{row['note']}: ₹{row['amount']}")
        total += int(row["amount"])

with open("data/config.json", "r", encoding="utf-8") as f:
    limit = json.load(f)["max_daily_limit"]

logging.info("Day total ₹%s (limit ₹%s)", total, limit)
print("Day total ₹%s (limit ₹%s)" % (total, limit))
print("Check logs/app.log")
```

Expected Output:

```
Tea: ₹120
Metro: ₹50
Day total ₹170 (limit ₹5000)
Check logs/app.log
```

Then open the kit (lab / tonight): `01_python_expense_tracker` — add expenses, export CSV, log to a file. Type it — do not paste `solution/`. **Also use:** 15_generator_csv_stream — now wire `yield` to `csv.DictReader` (stretch).

Practice Exercises

1. MedCare stores a short ward note in `ward.txt`. Write `"Bed 12: stable\n"` with `w`, then **append** `"Bed 14: observation\n"` with `a`. Read and print the file. If you used `w` the second time, what would be missing?

Model answer:

```
with open("ward.txt", "w", encoding="utf-8") as f:
    f.write("Bed 12: stable\n")
with open("ward.txt", "a", encoding="utf-8") as f:
    f.write("Bed 14: observation\n")
with open("ward.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

A second `"w"` would replace the file — Bed 12 would be gone.

2. Copy `source.txt` to `backup.txt` using `with open` on both files.

Model answer:

```
with open("source.txt", encoding="utf-8") as src, open("backup.txt", "w", encoding="utf-8") as dst:
    dst.write(src.read())
```

3. TechMart has `products.txt` (one product name per line). Count lines **without** calling `readlines()`. Print the count.

Model answer:

```
count = 0
with open("products.txt", encoding="utf-8") as f:
    for line in f:
        if line.strip():
            count += 1
print(count)
```

4. Read `appointments.csv` with columns `patient` and `fee`. Print the total fee. If the file is missing, print a clear message instead of crashing.

Model answer:

```
import csv

try:
    total = 0
    with open("appointments.csv", newline="", encoding="utf-8") as f:
        for row in csv.DictReader(f):
            total += int(row["fee"])
    print("Total fees:", total)
except FileNotFoundError:
    print("appointments.csv not found")
```

5. Load or create TechMart preferences in `prefs.json`. If the file exists, load it; otherwise start from `{"theme": "light", "lang": "en"}`. Set `theme` to `"dark"` and save.

Model answer:

```
import json
from pathlib import Path

path = Path("prefs.json")
prefs = {"theme": "light", "lang": "en"}
try:
    prefs = json.loads(path.read_text(encoding="utf-8"))
except FileNotFoundError:
    pass
prefs["theme"] = "dark"
path.write_text(json.dumps(prefs, indent=2), encoding="utf-8")
```

6. Configure a logger named `atm` that writes INFO+ to `atm.log` and WARNING+ to the console. Log one INFO and one WARNING.

Model answer:

```
import logging

logger = logging.getLogger("atm")
logger.setLevel(logging.INFO)
logger.handlers.clear()
fmt = logging.Formatter("%(levelname)s %(message)s")
fh = logging.FileHandler("atm.log", encoding="utf-8")
fh.setLevel(logging.INFO)
ch = logging.StreamHandler()
ch.setLevel(logging.WARNING)
fh.setFormatter(fmt)
ch.setFormatter(fmt)
logger.addHandler(fh)
logger.addHandler(ch)
logger.info("ATM session started")
logger.warning("Cash drawer low")
```

Mini-Project (Assignment): AcmeHR Expense Tracker with File + Logging

Title: DailyExpenseLogger
Customer: AcmeHR finance. Clerks add daily expenses. Closing Python must not wipe the month. A bad CSV row must not dump a traceback on the desk screen.
Lab kit (build here): 15_Lab_Assets/01_python_expense_tracker/
Also use: 15_generator_csv_stream — now wire `yield` to `csv.DictReader` (Files-week stretch)

This is **not** the in-class slice above. Same product and tools (`with`, CSV, JSON, logging), fuller ticket.

What to build

Piece	Detail
`data/expenses.csv`	Append: date, category, amount, note
`data/config.json`	Categories + daily limit
`logs/app.log`	All operations via `logging`
Summary	Total by category from CSV
Resilience	Missing file / bad CSV row → clear message, no crash

Folder layout: `main.py`, `data/config.json`, `data/expenses.csv`, `logs/app.log`

Stretch

- Enforce daily limit from config (warn or block when exceeded).
- Add a “export month report” that writes a new CSV.
- Stream a large seed file with a generator instead of `readlines()`.

Done when

- [ ] Starter kit / your CLI can add expense and show category totals
- [ ] Logging writes to a log file (not only `print`)
- [ ] Bad/missing files handled without traceback for expected cases
- [ ] Pushed to GitHub with sample `config.json` (no secrets)

Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. Difference between `read()` and iterating the file object?
A: `read()` loads the entire file into memory. Iteration reads one line at a time — constant memory for large files.

Q2. Purpose of the logging module vs `print`?
A: Logging provides levels, timestamps, handlers (file/console), configurable format, and production-ready filtering. `print` is ad-hoc debugging only. On a server, nobody is watching the terminal.

Q3. What is a context manager?
A: An object used with `with` that defines `__enter__` and `__exit__` for setup/cleanup. `open()` closes the file in `__exit__`, even if an exception occurred.

Q4. CSV vs JSON — how do you choose?
A: If the data is a **table** (Excel, import/export, reports) → CSV. If it is **config** or a nested structure (APIs, preferences) → JSON.

Q5. How do you handle a missing file?
A: `try` / `except FileNotFoundError`, or `pathlib.Path.exists()` / `os.path.isfile` first; return a default or prompt the user.

Q6. Difference between `json.load` and `json.loads`?
A: `json.load` reads from a file-like object. `json.loads` parses a JSON **string** already in memory (`s` for string).

Q7. Why is `with open` preferred over manual `close()`?
A: The context manager closes the file even if an exception occurs, avoiding resource leaks and ensuring data is flushed. Manual `close()` is easy to skip on error paths.

Q8. When use CSV vs JSON vs a database?
A: CSV suits flat tabular exports and spreadsheets. JSON fits nested config/API payloads. A database is better for concurrent writes, queries, relationships, and integrity in multi-user apps.

Q9. `tell()` vs `seek()`?
A: `tell()` returns the current cursor index (0-based). `seek(offset)` moves the cursor. Text mode in Python 3 seeks from the start; use a value from `tell()` if the offset is not 0.

Q10. `listdir()` vs `walk()`?
A: `listdir` lists one directory (not nested contents). `walk` visits that directory and all subdirectories, yielding path, subdir names, and file names.

Q11. What is pickling? When would you not use it?

A: Pickling serializes a Python object to bytes (`pickle.dump` / `load`). Do not unpickle untrusted files. For data you share with other languages or humans, prefer JSON or CSV.

Q12. Why is append mode safer than read-modify-write for log files?

A: Append adds at the end without loading or rewriting the whole file, so a crash is less likely to wipe earlier lines. Read-modify-write can truncate or corrupt if the rewrite fails halfway.

Q13. Binary vs text mode — when does each matter?

A: Text mode decodes characters (`encoding="utf-8"`) and translates newlines. Binary (`rb` / `wb`) is for images, zip, pickle, and any bytes you must not decode as text.

Q14. What sensitive data should never go in log files?

A: Passwords, OTPs, full card numbers, Aadhaar, session tokens — GDPR / India DPDP. Log that an auth attempt failed, not the secret itself.

Q15. How would you rotate log files when they grow too large?

A: Use `logging.handlers.RotatingFileHandler` (size) or `TimedRotatingFileHandler` (time) so old logs are renamed and a fresh file starts.

Q16. What is the difference between mode `w` and method `write()`?

A: `w` is *how you opened* the file (create/replace; it truncates). `write()` is *the call* that sends a string. Opening with `w` does not write data by itself — but it has already cleared the file.
